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Student Name:……

SPECIALIST MATHEMATICS UNITS 3 & 4

TRIAL EXAMINATION 2

2025

Reading Time: 15 minutes
Writing time: 2 hours

Instructions to students

This exam consists of Section A and Section B.
Section A consists of 20 multiple-choice questions and should be answered on the detachable answer sheet which can be found on page 26 of this exam.
Section B consists of 6 extended-answer questions.
Section A begins on page 2 of this exam and is worth 20 marks.
Section B begins on page 10 of this exam and is worth 60 marks.
There is a total of 80 marks available.
All questions in Section A and B should be answered.
In Section B, where more than one mark is allocated to a question, appropriate working must be shown.
An exact value is required to a question unless otherwise directed.
Unless otherwise stated, diagrams in this exam are not drawn to scale.
Students may bring one bound reference into the exam.
Students may bring into the exam one approved technology (calculator or software) and, if desired, one scientific calculator. Calculator memory does not need to be cleared. For approved computer-based CAS, full functionality may be used.
A formula sheet can be found at the end of this exam.

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SECTION A – Multiple-choice questions

Question 1

Consider the following claim:

For all $m \in R$, $m^2 \geq m$.

A value of m that can be used to give a counterexample to the above claim is

A. -2
B. 0
C. $\frac{1}{2}$
D. $\frac{3}{2}$

Question 2

Consider the following statement:

For all $m, n \in Z^+$, if m and n are odd, then mn is odd

Which of the following is the converse of the statement above?

A. if mn is even, then m and n are even
B. if mn is odd, then m and n are odd
C. if m and n are even, then mn is even
D. if mn is odd, then m and n are even

Question 3

The graph of $f(x) = \dfrac{\log_e(x^2)}{x-4}$ does **not** have

A. exactly one horizontal asymptote
B. exactly two vertical asymptotes
C. exactly one minimum stationary point
D. exactly one point of inflection

Question 4

When using a suitable substitution, $\int_0^{2\pi} \sin^3\left(\frac{x}{8}\right) dx$ can be expressed as

A. $8\int_1^{\frac{\sqrt{2}}{2}} u^2 - 1\, du$

B. $-8\int_0^{2\pi} 1 - u^2\, du$

C. $-8\int_1^{\frac{\sqrt{2}}{2}} u^3\, du$

D. $\frac{1}{8}\int_1^{\frac{\sqrt{2}}{2}} u^2 - 1\, du$

Question 5

Consider the three lines with the following vector equations:

$$\underset{\sim}{r}_1(t) = -2\underset{\sim}{i} + \underset{\sim}{j} + 3\underset{\sim}{k} + t(i + 2\underset{\sim}{j} + 3\underset{\sim}{k}) \quad , t \in R$$
$$\underset{\sim}{r}_2(s) = -3\underset{\sim}{i} + 2\underset{\sim}{j} + 6\underset{\sim}{k} + s(-2\underset{\sim}{i} + \underset{\sim}{j} + 4\underset{\sim}{k}) \quad , s \in R$$
$$\underset{\sim}{r}_3(u) = \underset{\sim}{i} + 4\underset{\sim}{j} - \underset{\sim}{k} + u(2\underset{\sim}{i} + \frac{13}{7}\underset{\sim}{j} + c\underset{\sim}{k}) \quad , u \in R$$

The value of $c \in R$, such that the three lines are concurrent, is

A. $\frac{1}{5}$

B. $-\frac{3}{5}$

C. $\frac{13}{7}$

D. $-\frac{23}{7}$

Question 6

For two vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$, the cross product $(\underset{\sim}{a} + \underset{\sim}{b}) \times (\underset{\sim}{a} - \underset{\sim}{b})$ simplifies to

A. $2(\underset{\sim}{a} \times \underset{\sim}{b})$

B. $2(\underset{\sim}{b} \times \underset{\sim}{a})$

C. $\underset{\sim}{0}$

D. 0

Question 7

The vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$ are linearly independent. If $(2x-3)\underset{\sim}{a}+(3y+8)\underset{\sim}{b}=\underset{\sim}{0}$, then the possible values of x and y are

A. $x \in R$ and $y \in R$

B. $x \in R$ and $y \in R$ provided $x = y$

C. $x = R \setminus \left\{\frac{3}{2}\right\}$ and $y = R \setminus \left\{-\frac{8}{3}\right\}$

D. $x = \frac{3}{2}$ and $y = -\frac{8}{3}$

Question 8

The scalar resolute of $\underset{\sim}{a} = -\underset{\sim}{i} + 4\underset{\sim}{j} + 2\underset{\sim}{k}$ in the direction of $\underset{\sim}{b} = \underset{\sim}{i} - 9\underset{\sim}{j} - 2\underset{\sim}{k}$ is

A. $\frac{41\sqrt{86}}{86}$

B. $\frac{41\sqrt{21}}{21}$

C. $-\frac{41\sqrt{21}}{21}$

D. $-\frac{41\sqrt{86}}{86}$

Question 9

If $z = b - bi$, $b \in R^+$, then the complex number z^n, $n \in Z$, is purely imaginary for

A. $n = 4k + 2$, $k \in Z$

B. $n = 2k + 2$, $k \in Z$

C. $n = 4k$, $k \in Z$

D. $n = 2k$, $k \in Z$

Question 10

The rays $\text{Arg}(z) = \frac{\pi}{3}$ and $\text{Arg}(z-1) = \frac{2\pi}{3}$, $z \in C$, intersect at the point

A. $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

B. $(0,0)$

C. $\left(\frac{1}{4}, \frac{\sqrt{3}}{4}\right)$

D. $(1,0)$

Question 11

Let $z = x + yi$, where $x, y \in R \setminus \{0\}$.

If $z - \frac{1}{\overline{z}}$, where $\overline{z}$ is the conjugate of z, is purely imaginary, which of the following must be true?

A. $x = y$

B. $|z| = 1$

C. $x = -y$

D. $z^2 = 1$

Question 12

A differential equation that has $y = x\cos(2x)$ as a solution is

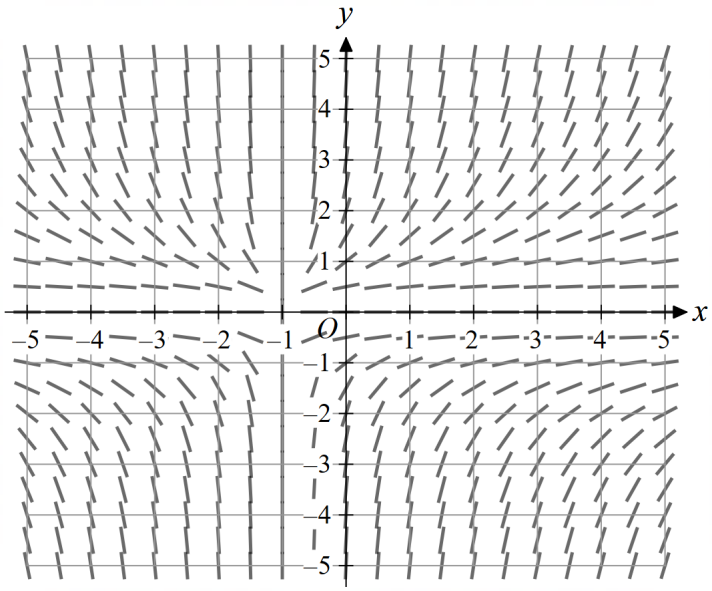
A. $\frac{d^2y}{dx^2} + y = -4\cos(2x)$

B. $\frac{d^2y}{dx^2} + y = -4\sin(2x)$

C. $\frac{d^2y}{dx^2} + 4y = -4\sin(2x)$

D. $\frac{d^2y}{dx^2} + 4y = 4\sin(2x)$

Question 13



The differential equation that has the diagram above as its directional field is

A. $\dfrac{dy}{dx} = \dfrac{y^2}{x+1}$

B. $\dfrac{dy}{dx} = \dfrac{x^2}{y+1}$

C. $\dfrac{dy}{dx} = \dfrac{y}{x+1}$

D. $\dfrac{dy}{dx} = xy$

Question 14

Consider $\dfrac{dy}{dx} = \dfrac{\log_e(x)}{y}$, where $y(1) = y_0 = 1$.

Using Euler's method with a step size of 0.1, an approximation to $y(0.8) = y_2$ is given by

A. 0.978
B. 0.989
C. 1.011
D. 1.022

Question 15

The position vector of a particle at time $t \geq 0$ is given by $\underset{\sim}{r}(t) = 3\sin(2t)\underset{\sim}{i} + 4\cos(4t)\underset{\sim}{j}$.
The path of the particle has a Cartesian equation of

A. $y = 8 - \dfrac{8x^2}{9}$

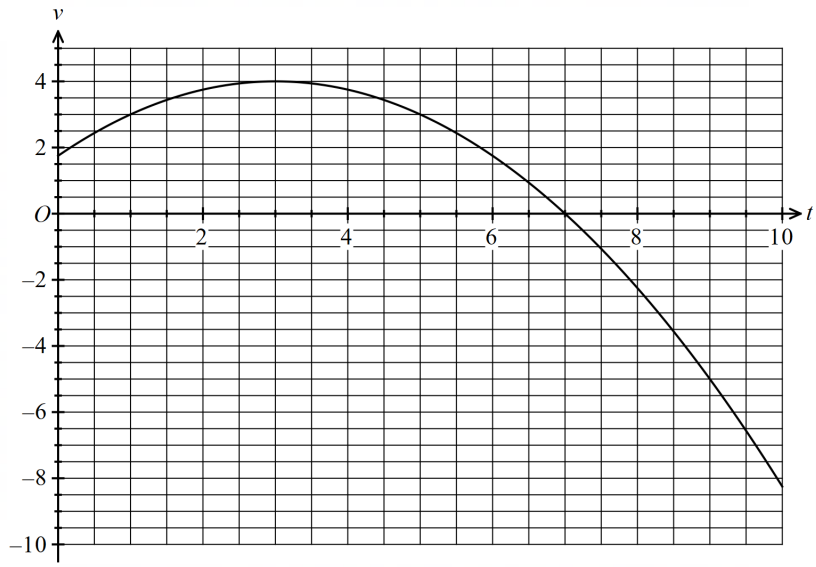
B. $y = 4 - \dfrac{8x^2}{9}$

C. $\dfrac{x^2}{9} + \dfrac{y^2}{16} = 1$

D. $y = 4 - \dfrac{4x^2}{9}$

Question 16

The velocity-time graph for a body moving along a straight line with a velocity v ms^{-1} after t seconds is shown below.



The velocity over the time interval $t \in [0,10]$ is given by $v(t) = \dfrac{-1}{4}(t-3)^2 + 4$.

The distance, in metres, travelled by the body over the ten seconds is closest to

A. 9.2
B. 11.3
C. 20.4
D. 31.7

Question 17

A function has the rule $f(x) = a\tan^{-1}(x-b)+c$, where $a, b \in R^+$ and $c \in R$. The graph of $y = f(x)$ will have one x-intercept if

A. $c \le -\dfrac{\pi a}{2}$

B. $c \ge \dfrac{\pi a}{2}$

C. $c \in \left(-\dfrac{\pi a}{2}, \dfrac{\pi a}{2}\right)$

D. $c \in \left[-\dfrac{\pi a}{2}, \dfrac{\pi a}{2}\right]$

Question 18

$3\sin(3\theta)\cos(2\theta)$ is equivalent to

A. $\dfrac{3}{2}\left(\sin(5\theta)+\sin(\theta)\right)$

B. $\dfrac{3}{2}\left(\cos(\theta)-\cos(5\theta)\right)$

C. $\dfrac{3}{2}\left(\cos(\theta)+\cos(5\theta)\right)$

D. $\dfrac{3}{2}\left(\sin(5\theta)-\sin(\theta)\right)$

Question 19

A pharmaceutical company is testing a new drug to see if it is more effective than the current standard treatment for a certain disease. The null hypothesis is that the new drug has the same effect as the current treatment (i.e., there is no difference between the two treatments). The alternative hypothesis is that the new drug is more effective than the current treatment.

A Type I error in this situation would be

A. Concluding that the new drug is no more effective than the old drug.
B. Concluding that the new drug is no more effective when in fact it is.
C. Concluding that the new drug is more effective when in fact it is not.
D. Concluding that the new drug is less effective when in fact it is more effective.

Question 20

The average time to wait at an ice cream store before being served has a mean μ minutes and a standard deviation σ minutes. A 95% confidence interval for the average customer wait time calculated from a sample of wait times for n customers is (4.55, 6.43). Using the same sample, a 99% confidence interval, correct to two decimal places, is

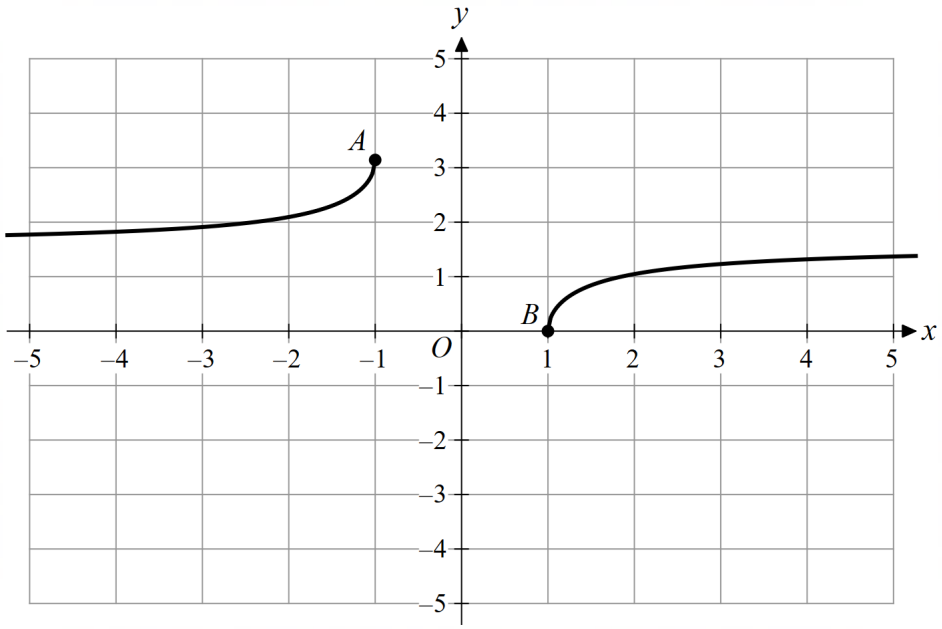
A. (4.50, 6.48)
B. (4.31, 6.79)
C. (4.25, 6.73)
D. cannot be determined without knowing the sample size

SECTION B

Question 1 (11 marks)

Consider the function $f(x) = \cos^{-1}\left(\frac{1}{x}\right)$ over its maximal domain.

The graph of $y = f(x)$ is shown below.



a. On the graph above, label the coordinates of points A and B, and draw in the horizontal asymptote, stating its equation. 2 marks

b. Hence or otherwise, state the domain and range of f. 1 mark

c. **i.** Show that the derivative of $f(x)$ is given by $f'(x) = \dfrac{1}{|x|\sqrt{x^2-1}}$. 2 marks

ii. Hence, state the domain and range of $f'(x)$. 2 marks

Consider the function $g(x) = \cos^{-1}\left(\frac{ax+1}{x}\right)$, where $a \in R$.

d. State the value(s) of a, such that:

i. $g(x)$ has a horizontal asymptote. 1 mark

ii. the graph of $g(x)$ is only defined for positive x values. 1 mark

e. State the possible value(s) of a, correct to three decimal places, such that the distance between the endpoints of the graph of $g(x)$ is 5 units. 2 marks

Question 2 (10 marks)

Let U be the subset of the complex plane given by

$$U = \{z : |z| = 4, z \in C\}$$

a. Describe geometrically what the subset U represents. 1 mark

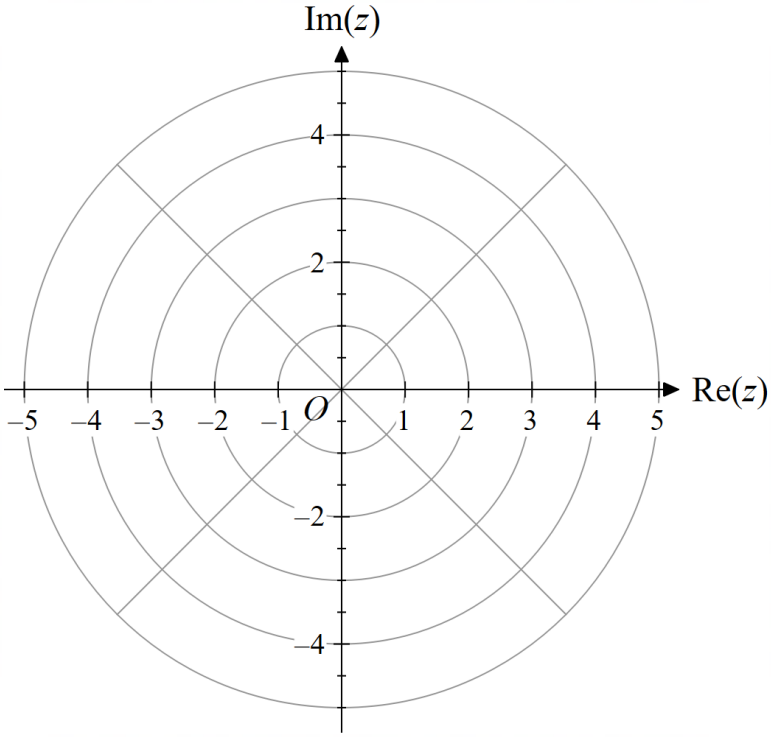
Let the line L have a Cartesian equation $y = \frac{-\sqrt{3}}{3}x$ and domain $x < 0$. L can also be defined as $\text{Arg}(z) = \alpha$, $z \in C$.

b. State the value of α. 1 mark

L can also be defined as $|z + 1 - \sqrt{3}i| = |z - \beta|$, $z \in C$.

c. Find β. 1 mark

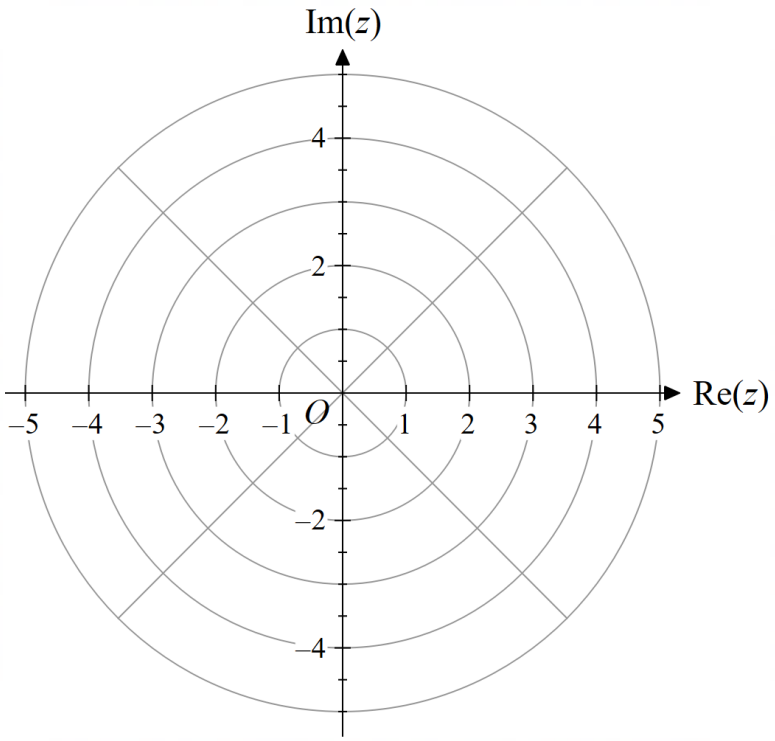
d. On the Argand diagram below, sketch the subset U and the sketch the line L. 2 marks



e. Find the area bounded by the line L, the graph of $|z| \le 4$, the real axis and the vertical line where $\text{Re}(z) = -3$. 2 marks

Let $W = \left\{ z : 2 < |z| < 4 \text{ and } 0 < \text{Arg}(z) \le \frac{\pi}{4}, z \in C \right\}$.

f. Sketch W on the Argand diagram below. 2 marks



g. Find $z \in W$, where $z = a + bi$, and $a, b \in Z$. 1 mark

Question 3 (10 marks)

Consider the curve defined by the following parametric equations:

$$x = t^2 + 4$$
$$y = 3t \qquad , \ t \in R$$

All lengths are measured in centimetres.

a. Find the Cartesian equation of the curve and state its domain. 2 marks

b. Find the length of the curve for $t \in [0, 4]$, correct to two decimal places. 1 mark

The curve is rotated for $t \in [0, 4]$ about the y-axis to form a solid of revolution.

c. Find the total surface area of the solid formed, correct to two decimal places. 2 marks

The curve rotated for $t \in [0,4]$ about the y-axis is to be made into a glass with a height of 12 cm. Water is then poured into the initially empty glass to a height of h cm.

d. Find the volume of water, in cm^3, in the glass at height h cm. 1 mark

e. If the water is poured into the empty glass at a rate of 400 cm^3/second, at what rate is the height of the water increasing when $h = 5$ cm?
Give your answer in units of cm/sec correct to two decimal places. 2 marks

Instead, the water is poured into the empty glass at a rate of $200r$ cm^3/second, where r is the time in seconds from when water begins to be poured in.

f. **i.** Write a rule for the time, r seconds, it takes to fill the glass to a height of h cm. 1 mark

ii. Hence, or otherwise, determine how long it takes to fill the glass. Give your answer in seconds correct to two decimal places. 1 mark

Question 4 (9 marks)

A town experiences an outbreak of a contagious virus. The number of people infected by the virus t days after the outbreak begins is denoted by N.

The spread of the virus is modelled by the logistic differential equation:

$$\frac{dN}{dt} = 0.05N\left(4 - \frac{N}{2500}\right)$$

a. What is the maximum number of infected people in the town? 1 mark

Initially the number of infected people is 10.

b. Setup and solve a definite integral to find how long it takes for 70 people to be infected by the virus. Give your answer correct to two decimal places. 1 mark

c. **i.** Find the rule for $N(t)$ in terms of t. Give your answer in the form

$\dfrac{ae^{\frac{t}{5}}}{e^{\frac{t}{5}}+b}$, where $a, b \in R$. 2 marks

ii. Hence, or otherwise, find after how many days the disease is spreading most rapidly. Give your answer correct to three decimal places. 1 mark

Health officials start testing infected individuals once the number of infected people exceeds 1000. The number of people being tested per day, T, is directly proportional to the rate at which the infection is spreading.

d. **i.** If 200 people are tested per day when 2000 people are infected, show that the constant of proportionality is 0.625. 1 mark

ii. Hence, express T in terms of N. 1 mark

e. After how many days since testing begins is the number of tests per day a maximum? Give your answer correct to two decimal places. 2 marks

Question 5 (10 marks)

Consider the points $A(1,2,4)$, $B(2,-1,3)$ and $C(3,-2,6)$.

a. **i.** Find the vector $\overrightarrow{AB} \times \overrightarrow{AC}$. 2 marks

ii. Show that the Cartesian equation of the plane Π_1 that contains the points A, B and C is $5x + 2y - z = 5$. 1 mark

A second plane, Π_2, has the Cartesian equation $2x - y - z = 10$.

b. Find the acute angle between the planes Π_1 and Π_2, in degrees, correct to two decimal places. 1 mark

c. Let L be the line of intersection of the planes Π_1 and Π_2.

i. Find the coordinates of the point D, where L crosses the x-y plane. 2 marks

ii. A vector equation of the line L is of the form
$\underset{\sim}{L}(\lambda) = x\underset{\sim}{i} + y\underset{\sim}{j} + \lambda\left(\underset{\sim}{i} - \underset{\sim}{j} + 3\underset{\sim}{k}\right)$, $x, y \in R$. State the values of x and y. 1 mark

d. A third plane, Π_3, has the Cartesian equation $x + y + 4z = 11$. Find:

i. the angle between the line L and the plane Π_3, in degrees correct to two decimal places. 1 mark

ii. the point of intersection between the line L and Π_3. 2 marks

Question 6 (10 marks)

The time it takes for a car to drive through a small country town is a continuous random variable X with probability density function:

$$f(x)=\begin{cases}\dfrac{-9}{2450}(x-3)(x-11)+0.1 & , \ x\in[3,10]\\ 0 & , \text{ otherwise}\end{cases}$$

a. State definite integrals for the mean $\mathrm{E}(X)$ and variance $\mathrm{Var}(X)$, that when solved correct to three decimal places, give $\mathrm{E}(X)=6.605$ and $\mathrm{Var}(X)=3.729$. 2 marks

b. Using the values given in part **a.**, determine for a random sample of 50 cars the probability that their average time driving through the town is more than seven minutes. Give your answer correct to four decimal places. 2 marks

People who decide to stop in the small country town often visit the bakery. The total time people wait in the bakery is normally distributed with a mean of 12 minutes and a standard deviation of 2.5 minutes. Assume the time a person waits in the bakery is independent from any other person.

c. Determine the probability that a randomly selected person who visits the bakery must wait more than 13 minutes. Give your answer correct to four decimal places. 1 mark

d. What is the probability that, for the next 10 customers that visit the bakery, their total time waiting in the bakery is more than 135 minutes? Give your answer correct to four decimal places. 2 marks

Next door to the bakery is a café. The total time people wait in the café collecting their takeaway food is normally distributed with a mean of 11 minutes and a standard deviation of 1.5 minutes.

e. If Brett enters the **bakery** at the same time as Claire enters the **café**, determine the probability that Brett exits the bakery before Claire exits the café. Give your answer correct to four decimal places. 2 marks

f. How much earlier than Claire entering the café should Brett enter the bakery, if he wants to be 90% sure he exits the bakery before Claire exits the café? Give your answer in minutes correct to two decimal places. 1 mark

End of Examination

Specialist Mathematics Formula Sheet

Mensuration

area of a circle segment	$\frac{r^2}{2}(\theta - \sin(\theta))$	volume of a sphere	$\frac{4}{3}\pi r^3$
volume of a cylinder	$\pi r^2 h$	area of a triangle	$\frac{1}{2}bc\sin(A)$
volume of a cone	$\frac{1}{3}\pi r^2 h$	sine rule	$\frac{a}{\sin(A)} = \frac{b}{\sin(B)} = \frac{c}{\sin(C)}$
volume of a pyramid	$\frac{1}{3}Ah$	cosine rule	$c^2 = a^2 + b^2 - 2ab\cos(C)$

Algebra, number and structure (complex numbers)

$z = x + iy = r(\cos(\theta) + i\sin(\theta)) = r\text{cis}(\theta)$	$\lvert z\rvert = \sqrt{x^2 + y^2} = r$	
$-\pi < \text{Arg}(z) \le \pi$	$z_1 z_2 = r_1 r_2 \text{cis}(\theta_1 + \theta_2)$	
$\frac{z_1}{z_2} = \frac{r_1}{r_2}\text{cis}(\theta_1 - \theta_2)$	de Moivre's theorem	$z^n = r^n \text{cis}(n\theta)$

Data analysis, probability and statistics

for independent random variables $X_1, X_2 \ldots X_n$	$\mathrm{E}(aX_1 + b) = a\,\mathrm{E}(X_1) + b$ $\mathrm{E}(a_1X_1 + a_2X_2 + \ldots + a_nX_n)$ $= a_1\mathrm{E}(X_1) + a_2\mathrm{E}(X_2) + \ldots + a_n\mathrm{E}(X_n)$	
	$\mathrm{Var}(aX_1 + b) = a^2\mathrm{Var}(X_1)$ $\mathrm{Var}(a_1X_1 + a_2X_2 + \ldots + a_nX_n)$ $= a_1^2\mathrm{Var}(X_1) + a_2^2\mathrm{Var}(X_2) + \ldots + a_n^2\mathrm{Var}(X_n)$	
for independent identically distributed variables $X_1, X_2 \ldots X_n$	$\mathrm{E}(X_1 + X_2 + \ldots + X_n) = n\mu$	
	$\mathrm{Var}(X_1 + X_2 + \ldots + X_n) = n\sigma^2$	
approximate confidence interval for μ	$\left(\bar{x} - z\frac{s}{\sqrt{n}},\ \bar{x} + z\frac{s}{\sqrt{n}}\right)$	
distribution of sample mean $\bar{X}$	mean	$\mathrm{E}(\bar{X}) = \mu$
	variance	$\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}$

Calculus

$\frac{d}{dx}\left(x^n\right) = nx^{n-1}$	$\int x^n\,dx = \frac{1}{n+1}x^{n+1} + c,\ n \neq -1$
$\frac{d}{dx}\left(e^{ax}\right) = ae^{ax}$	$\int e^{ax}\,dx = \frac{1}{a}e^{ax} + c$
$\frac{d}{dx}\left(\log_e(x)\right) = \frac{1}{x}$	$\int \frac{1}{x}\,dx = \log_e\lvert x\rvert + c$
$\frac{d}{dx}\left(\sin(ax)\right) = a\cos(ax)$	$\int \sin(ax)\,dx = -\frac{1}{a}\cos(ax) + c$
$\frac{d}{dx}\left(\cos(ax)\right) = -a\sin(ax)$	$\int \cos(ax)\,dx = \frac{1}{a}\sin(ax) + c$
$\frac{d}{dx}\left(\tan(ax)\right) = a\sec^2(ax)$	$\int \sec^2(ax)\,dx = \frac{1}{a}\tan(ax) + c$
$\frac{d}{dx}\left(\cot(ax)\right) = -a\operatorname{cosec}^2(ax)$	$\int \operatorname{cosec}^2(ax)\,dx = -\frac{1}{a}\cot(ax) + c$
$\frac{d}{dx}\left(\sec(ax)\right) = a\sec(ax)\tan(ax)$	$\int \sec(ax)\tan(ax)\,dx = \frac{1}{a}\sec(ax) + c$
$\frac{d}{dx}(\operatorname{cosec}(ax)) = -a\operatorname{cosec}(ax)\cot(ax)$	$\int \operatorname{cosec}(ax)\cot(ax)\,dx = -\frac{1}{a}\operatorname{cosec}(ax) + c$
$\frac{d}{dx}\left(\sin^{-1}(ax)\right) = \frac{a}{\sqrt{1-(ax)^2}}$	$\int \frac{1}{\sqrt{a^2-x^2}}\,dx = \sin^{-1}\left(\frac{x}{a}\right) + c,\ a > 0$
$\frac{d}{dx}\left(\cos^{-1}(ax)\right) = \frac{-a}{\sqrt{1-(ax)^2}}$	$\int \frac{-1}{\sqrt{a^2-x^2}}\,dx = \cos^{-1}\left(\frac{x}{a}\right) + c,\ a > 0$
$\frac{d}{dx}\left(\tan^{-1}(ax)\right) = \frac{a}{1+(ax)^2}$	$\int \frac{a}{a^2+x^2}\,dx = \tan^{-1}\left(\frac{x}{a}\right) + c$
	$\int (ax+b)^n\,dx = \frac{1}{a(n+1)}(ax+b)^{n+1} + c,\ n \neq -1$
	$\int \frac{1}{ax+b}\,dx = \frac{1}{a}\log_e\lvert ax+b\rvert + c$

Calculus - continued

product rule	$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$
quotient rule	$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$
chain rule	$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$
integration by parts	$\int u\frac{dv}{dx}dx = uv - \int v\frac{du}{dx}dx$
Euler's method	If $\frac{dy}{dx} = f(x, y), x_0 = a$ and $y_0 = b$, then $x_{n+1} = x_n + h$ and $y_{n+1} = y_n + h \times f(x_n, y_n)$
arc length parametric	$\int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$
surface area Cartesian about x-axis	$\int_{x_1}^{x_2} 2\pi y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$
surface area Cartesian about y-axis	$\int_{y_1}^{y_2} 2\pi x\sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$
surface area parametric about x-axis	$\int_{t_1}^{t_2} 2\pi y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$
surface area parametric about y-axis	$\int_{t_1}^{t_2} 2\pi x\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$

Kinematics

acceleration	$a = \frac{d^2x}{dt^2} = \frac{dv}{dt} = v\frac{dv}{dx} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$	
constant acceleration formulas	$v = u + at$	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$	$s = \frac{1}{2}(u+v)t$

Vectors in two and three dimensions

$\underset{\sim}{r}(t) = x(t)\underset{\sim}{i} + y(t)\underset{\sim}{j} + z(t)\underset{\sim}{k}$	$\lvert\underset{\sim}{r}(t)\rvert = \sqrt{x(t)^2 + y(t)^2 + z(t)^2}$
	$\dot{\underset{\sim}{r}}(t) = \frac{d\underset{\sim}{r}}{dt} = \frac{dx}{dt}\underset{\sim}{i} + \frac{dy}{dt}\underset{\sim}{j} + \frac{dz}{dt}\underset{\sim}{k}$
for $\underset{\sim}{r}_1 = x_1\underset{\sim}{i} + y_1\underset{\sim}{j} + z_1\underset{\sim}{k}$ and $\underset{\sim}{r}_2 = x_2\underset{\sim}{i} + y_2\underset{\sim}{j} + z_2\underset{\sim}{k}$	vector scalar product $\underset{\sim}{r}_1 . \underset{\sim}{r}_2 = \lvert\underset{\sim}{r}_1\rvert\lvert\underset{\sim}{r}_2\rvert\cos(\theta) = x_1x_2 + y_1y_2 + z_1z_2$
	vector cross product $\underset{\sim}{r}_1 \times \underset{\sim}{r}_2 = \begin{vmatrix} \underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k} \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix} = (y_1z_2 - y_2z_1)\underset{\sim}{i} + (x_2z_1 - x_1z_2)\underset{\sim}{j} + (x_1y_2 - x_2y_1)\underset{\sim}{k}$
vector equation of a line	$\underset{\sim}{r}(t) = \underset{\sim}{r}_1 + t\underset{\sim}{r}_2 = (x_1 + x_2t)\underset{\sim}{i} + (y_1 + y_2t)\underset{\sim}{j} + (z_1 + z_2t)\underset{\sim}{k}$
parametric equation of a line	$x(t) = x_1 + x_2t \qquad y(t) = y_1 + y_2t \qquad z(t) = z_1 + z_2t$
vector equation of a plane	$\underset{\sim}{r}(s,t) = \underset{\sim}{r}_0 + s\underset{\sim}{r}_1 + t\underset{\sim}{r}_2$ $= (x_0 + x_1s + x_2t)\underset{\sim}{i} + (y_0 + y_1s + y_2t)\underset{\sim}{j} + (z_0 + z_1s + z_2t)\underset{\sim}{k}$
parametric equation of a plane	$x(s,t) = x_0 + x_1s + x_2t, \quad y(s,t) = y_0 + y_1s + y_2t, \quad z(s,t) = z_0 + z_1s + z_2t$
Cartesian equation of a plane	$ax + by + cz = d$

Circular functions

$\cos^2(x)+\sin^2(x)=1$	
$1+\tan^2(x)=\sec^2(x)$	$\cot^2(x)+1=\operatorname{cosec}^2(x)$
$\sin(x+y)=\sin(x)\cos(y)+\cos(x)\sin(y)$	$\sin(x-y)=\sin(x)\cos(y)-\cos(x)\sin(y)$
$\cos(x+y)=\cos(x)\cos(y)-\sin(x)\sin(y)$	$\cos(x-y)=\cos(x)\cos(y)+\sin(x)\sin(y)$
$\tan(x+y)=\dfrac{\tan(x)+\tan(y)}{1-\tan(x)\tan(y)}$	$\tan(x-y)=\dfrac{\tan(x)-\tan(y)}{1+\tan(x)\tan(y)}$
$\sin(2x)=2\sin(x)\cos(x)$	
$\cos(2x)=\cos^2(x)-\sin^2(x)=2\cos^2(x)-1=1-2\sin^2(x)$	$\tan(2x)=\dfrac{2\tan(x)}{1-\tan^2(x)}$
$\sin^2(ax)=\dfrac{1}{2}\left(1-\cos(2ax)\right)$	$\cos^2(ax)=\dfrac{1}{2}\left(1+\cos(2ax)\right)$

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End of Formula Sheet

SPECIALIST MATHS UNITS 3&4

TRIAL EXAMINATION 2

MULTIPLE-CHOICE ANSWER SHEET

STUDENT NAME …….

INSTRUCTIONS

Fill in the letter that corresponds to your choice. Example: (A) ■ (C) (D)

The answer selected is B. Only one answer should be selected.

1.	(A)	(B)	(C)	(D)
2.	(A)	(B)	(C)	(D)
3.	(A)	(B)	(C)	(D)
4.	(A)	(B)	(C)	(D)
5.	(A)	(B)	(C)	(D)
6.	(A)	(B)	(C)	(D)
7.	(A)	(B)	(C)	(D)
8.	(A)	(B)	(C)	(D)
9.	(A)	(B)	(C)	(D)
10.	(A)	(B)	(C)	(D)
11.	(A)	(B)	(C)	(D)
12.	(A)	(B)	(C)	(D)
13.	(A)	(B)	(C)	(D)
14.	(A)	(B)	(C)	(D)
15.	(A)	(B)	(C)	(D)
16.	(A)	(B)	(C)	(D)
17.	(A)	(B)	(C)	(D)
18.	(A)	(B)	(C)	(D)
19.	(A)	(B)	(C)	(D)
20.	(A)	(B)	(C)	(D)